

CASE 40

MYASTHENIA GRAVIS

The immune response turns against the host.

The specific adaptive immune response can, in rare instances, be mounted against self antigens and cause autoimmune disease. Injury to body tissues can result from antibodies directed against cell-surface or extracellular-matrix molecules, from antibodies bound to circulating molecules that deposit as immune complexes, or from clones of T cells that react with self antigens. A special class of autoimmune disease is caused by autoantibodies against cell-surface receptors (Fig. 40.1). Graves' disease and myasthenia gravis are two well-studied examples. Graves' disease is caused by autoantibodies against the receptor on thyroid cells

Some common type II autoimmune diseases caused by antibody against surface or matrix antigens

Syndrome	Autoantigen	Consequence
Autoimmune hemolytic anemia (see Case 39)	Rh blood group antigens, I antigen	Destruction of red blood cells by complement and phagocytes, anemia
Autoimmune thrombocytopenic purpura	Platelet integrin GpIIb/IIIa	Abnormal bleeding
Goodpasture's syndrome	Noncollagenous domain of basement membrane collagen type IV	Glomerulonephritis Pulmonary hemorrhage
Pemphigus vulgaris (see Case 41)	Epidermal cadherin	Blistering of skin
Graves' disease	Thyroid-stimulating hormone receptor	Hyperthyroidism
Myasthenia gravis	Acetylcholine receptor	Progressive weakness
Insulin-resistant diabetes	Insulin receptor (antagonist)	Hyperglycemia, ketoacidosis
Hypoglycemia	Insulin receptor (agonist)	Hypoglycemia

Fig. 40.1 Autoimmune diseases caused by antibody against surface or matrix antigens. These are known as type II autoimmune diseases. Damage by IgE-mediated responses (type I) does not occur in autoimmune disease. In most type II diseases, autoantibodies bind to the cell surface or extracellular matrix and target them for destruction by phagocytes (often with the help of complement) and/or natural killer cells. A special class of autoimmune diseases is caused by autoantibodies that bind cellular receptors and either stimulate or block their normal function. Immune-complex disease (type III) is discussed in Case 36. T cell-mediated disease (type IV) is discussed in Cases 35, 38, and 49.

TOPICS BEARING ON THIS CASE:

Humoral autoimmunity

Transfer of maternal antibodies

Mechanisms for breaking tolerance

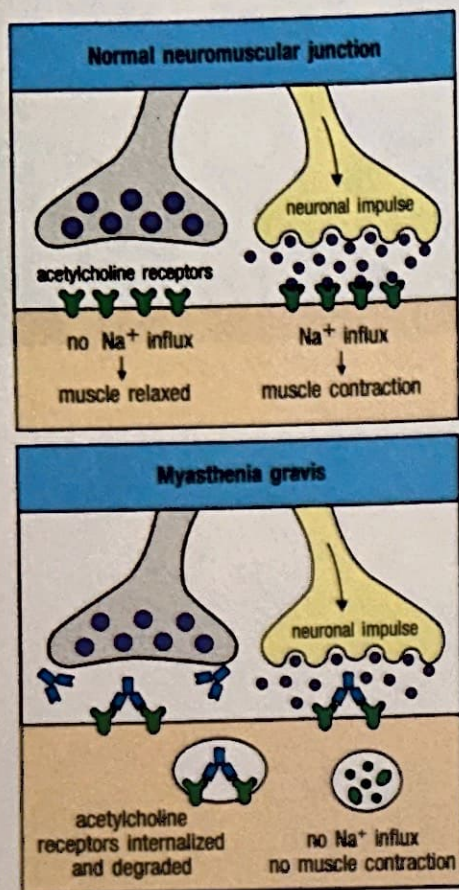


Fig. 40.2 Autoantibodies against the acetylcholine receptor weaken the reception of the signal from nerve ends that cause the muscle cell to contract. At the neuromuscular junction, acetylcholine is released from stimulated neurons and binds to acetylcholine receptors, triggering muscle contraction. The acetylcholine is destroyed rapidly by the enzyme acetylcholinesterase after release. In myasthenia gravis, autoantibodies against the acetylcholine receptor induce its endocytosis and degradation, and prevent muscles from responding to neuronal impulses.

for thyroid-stimulating hormone (TSH), secreted by the pituitary gland. In this disease, the autoantibody binds to the TSH receptor, and like TSH, it stimulates the thyroid gland to produce thyroid hormones. In myasthenia gravis, the opposite effect is observed: antibodies against the acetylcholine receptor at the neuromuscular junction impede the binding of acetylcholine and stimulate internalization of the receptor, thereby blocking the transmission of nerve impulses by acetylcholine (Fig. 40.2). In addition, the presence of autoantibodies at the neuromuscular junction initiates complement-mediated lysis of the muscle endplate and damages the muscle membrane.

Myasthenia gravis means severe (*gravis*) muscle (*my*) weakness (*asthenia*). The autoimmune nature of the disease was demonstrated in the early 1970s, when Patrick and Lindstrom showed that rabbits immunized with purified acetylcholine receptors developed antibodies against the acetylcholine receptor. Moreover, the rabbits developed floppy ears, which were reminiscent of the droopy eyelids (ptosis) that are the most characteristic symptom of myasthenia gravis in humans. Subsequently, patients with this disease were found to have antibodies against the acetylcholine receptor. In addition, pregnant women with myasthenia gravis may transfer the disease to their newborn infants. As IgG is the only maternal serum protein that crosses the placenta from mother to fetus, neonatal myasthenia gravis is clear evidence that myasthenia gravis is caused by an autoantibody of IgG isotype. More recently, patients with myasthenia gravis have been identified who have autoantibodies against muscle-specific kinase (MUSK) rather than the acetylcholine receptor. MUSK is a tyrosine kinase receptor involved in clustering acetylcholine receptors; therefore, these autoantibodies also inhibit signaling through the neuromuscular junction.

The case of Mr Weld: from floppy ears to droopy eyelids.

Mr Weld, a 71-year-old retired engineer, had been in good health and active all his life. He developed double vision (diplopia). Initially, he did not want to seek medical attention because the double vision sometimes improved spontaneously. However, it gradually worsened over the course of 4 months and he finally scheduled an appointment with his physician.

On examination, the doctor noticed that Mr Weld had ptosis of both eyelids so that they covered the upper third of the irises of his eyes. When the doctor asked Mr Weld to look to the right and then to the left, he noticed limitations in the ocular movements of both eyes, as shown in Fig. 40.3.

The remainder of the neurological examination was normal. No other muscle weakness was found during the examination.

A radiological examination of the chest was performed, and it was normal. There was no evidence in the radiograph of enlargement of the thymus gland. A blood sample was taken from Mr Weld, and his serum was tested for antibodies against the acetylcholine receptor. The serum contained 6.8 units of antibody against the

Ptosis of both eyelids.
Limitation in ocular
movements of both eyes.

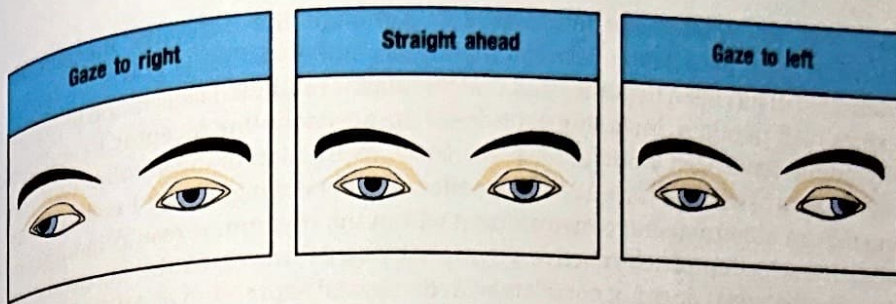


Fig. 40.3 Diagram of ocular movement limitation.

acetylcholine receptor (normal less than 0.5 units). Mr Weld was told to take pyridostigmine, an inhibitor of cholinesterase. His double vision improved steadily but he developed diarrhea from the pyridostigmine, and this limited the amount he could take.

Three years later, Mr Weld developed a severe respiratory infection. Soon afterward, his ptosis became so severe that he had to lift his eyelids by taping them with adhesive tape. His diplopia recurred and his speech became indistinct. He developed difficulty in chewing and swallowing food. He could only tolerate a diet of soft food and it would take him several hours to finish a meal.

On examination the neurologist noted that Mr Weld now had weakness of the facial muscles and the tongue, and the abnormality in ocular movements again became apparent. Because of the diarrhea Mr Weld was only able to tolerate one-quarter of the prescribed dose of pyridostigmine. He also developed difficulty in breathing. His vital capacity (the amount of air he could exhale in one deep breath) was low, at 3.5 liters.

He was admitted to hospital and treated with azathioprine. Thereafter he showed steady improvement. His ptosis and diplopia improved remarkably and he was able to eat normally. His vital capacity returned to normal and was measured to be 5.1 liters.

Prescribe pyridostigmine.

Symptoms recur during severe respiratory infection; difficulty in chewing and swallowing.

Myasthenia gravis

The defining characteristic of myasthenia gravis is a fluctuating weakness that worsens with activity and improves with rest. Normally, repetitive nerve stimulation during sustained physical activity results in the release of decreased amounts of acetylcholine with each successive stimulus; however, enough acetylcholine is released to achieve the desired muscle strength in healthy individuals. In contrast, patients with myasthenia gravis have fewer functional acetylcholine receptors as a result of the presence of anti-receptor autoantibodies. During repetitive nerve stimulation, the combination of fewer functional acetylcholine receptors with the physiologic decrease in neurotransmitter release results in muscular weakness.

Mr Weld experienced a common type of myasthenia gravis, called the oculobulbar form because it primarily affects the muscles of the eye. Older patients tend to have more generalized muscle weakness as well, and often have autoantibodies against muscle proteins in addition to anti-acetylcholine receptor antibodies. In very severe cases, difficulty in swallowing can cause the aspiration of food particles into the lung and impaired breathing, which may be fatal. Plasmapheresis (the filtration and removal of plasma from whole blood) can be used to remove the autoantibodies and treat a myasthenic crisis.

In younger people, the disease presents most often with weakness in the eye muscles. Chest radiographs of younger people with myasthenia gravis frequently reveal enlargement of the thymus gland; however, an association between myasthenia gravis and tumors of the thymus (thymomas) is more common in adults. Early removal of the thymus gland (thymectomy), particularly in those with thymomas,

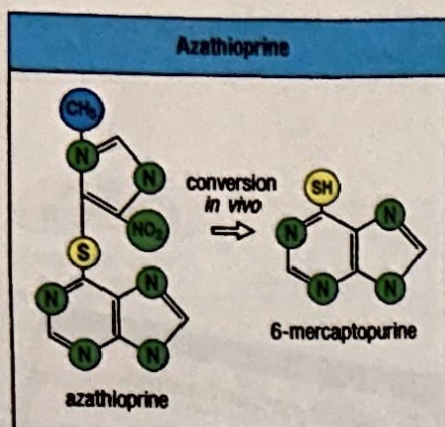


Fig. 40.4 The structure of azathioprine and its active product, mercaptopurine.

may lead to symptomatic improvement. Although the definitive mechanisms underlying the association between thymomas and myasthenia gravis are not yet identified, it has been hypothesized that neoplastic epithelial cells in the thymoma express self-peptides, including peptides of the acetylcholine receptor. In addition, thymomas have been found to have decreased expression of the autoimmune regulator gene (*AIRE*; see Case 17) and smaller numbers of regulatory T cells, indicating that an abnormal microenvironment within the thymoma results in impaired negative selection of self-reactive T lymphocytes. However, as the occurrence of myasthenia gravis does not correlate with decreased expression of *AIRE* in the thymus, there are still unidentified factors that influence the development of myasthenia gravis in patients with thymomas.

Questions.

- 1 Newborn infants of mothers with myasthenia gravis exhibit symptoms of myasthenia gravis at birth. How long would the disease be likely to last in these infants?
- 2 Pyridostigmine is an ideal drug for the treatment of myasthenia gravis. It inhibits the enzyme cholinesterase, which normally cleaves and inactivates acetylcholine. In this way, pyridostigmine prolongs the biological half-life of acetylcholine. Unfortunately, it also causes diarrhea by increasing the amount of acetylcholine in the intestine. Acetylcholine binds to the muscarinic receptors in the intestine and increases intestinal motility. Because he could not tolerate full therapeutic doses of pyridostigmine and was getting worse, Mr Weld was given azathioprine (Fig. 40.4) and showed marked improvement. What did the azathioprine do? What would concern you about prolonged use of this drug?
- 3 Mr Weld had a severe relapse in his disease after a respiratory infection. Many autoimmune diseases seem to be triggered by infection, and relapses in autoimmune diseases frequently follow an infection. Can you explain how this might happen?